| No. | Paper / Reference | Journal & Quartile | Dataset / Method | Research Gap | How It Supports the Study |
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| 1 | Awan, M. J., Masood, O. A., Mohammed, M. A., Yasin, A., Zain, A. M., Damaševičius, R., & Abdulkareem, K. H. (2021). Image-Based Malware Classification Using VGG19 Network and Spatial Convolutional Attention. Electronics, 10(19), 2444. | Electronics (MDPI) 2021 SJR 2024: Q2 | MalImg; VGG19 + Spatial Convolutional Attention (SACNN); experiments with and without class balancing. | The study demonstrates strong image-based classification, but the focus remains on CNN/attention performance within the MalImg benchmark. This leaves room for broader evaluation of class-specific errors, robustness beyond the benchmark, and alternative representations that explicitly model relationships among image regions. The paper's class-balancing experiments also motivate more detailed macro-level and per-family evaluation rather than relying mainly on aggregate performance. | Useful baseline for CNN + attention and class-imbalance analysis. A follow-up study can compare attention-based CNNs with alternative representations and report macro-F1, per-class recall, and confusion-matrix/error analysis. |
| 2 | Hammad, B. T., Jamil, N., Ahmed, I. T., Zain, Z. M., & Basheer, S. (2022). Robust Malware Family Classification Using Effective Features and Classifiers. Applied Sciences, 12(15), 7877. | Applied Sciences (MDPI) 2022 SJR 2024: Q2; JCR 2024: Q2 | MalImg; Tamura handcrafted features and GoogLeNet deep features; KNN, SVM, and ELM classifiers; unbalanced dataset. | The approach combines handcrafted and deep features with conventional classifiers and explicitly works with the imbalanced MalImg dataset. However, it does not explore newer end-to-end attention/transformer architectures or explainability of deep visual decisions. The reliance on engineered texture features also leaves an opportunity to test whether modern end-to-end models can learn more transferable representations without handcrafted feature | Provides a strong feature-engineering/classical-classifier comparison point. A newer study can benchmark against it using end-to-end CNN/attention models and stronger class-wise metrics. |

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| | | | | design. | |
| 3 | Al-Masri, B., Bakir, N., El-Zaart, A., & Samrouth, K. (2024). Dual Convolutional Malware Network (DCMN): An Image-Based Malware Classification Using Dual Convolutional Neural Networks. Electronics, 13(18), 3607. | Electronics (MDPI)<br>2024<br>SJR 2024: Q2 | MalImg; dual-CNN image-based malware classification. | The dual-CNN design extends conventional CNN classification, but the image representation still treats the malware visualization primarily as a regular image grid. This leaves an open opportunity to investigate whether explicit relationships among local image patches/regions can provide complementary information. More detailed robustness and failure analysis can also clarify which malware families remain difficult despite high overall performance. | Represents a recent CNN-based MalImg baseline. It motivates comparing ordinary convolutional feature extraction with models that explicitly capture long-range or region-to-region relationships. |
| 4 | Musaev, A., Anorboev, A., & Youn, J. M. (2025). Optimized Epoch Selection Ensemble: Integrating Custom CNN and Fine-Tuned MobileNetV2 for Malimg Dataset Classification. IEEE Access, 13, 45623–45633. | IEEE Access<br>2025<br>SJR 2024: Q1<br>Open access (CC BY 4.0) | MalImg; custom CNN + fine-tuned MobileNetV2; optimized ensemble member selection using true-prediction analysis. | The paper addresses ensemble-member selection and reports 99.05% accuracy, but its contribution is centered on combining CNN models and selecting useful training epochs. It does not directly investigate graph-based representations of malware images, and the benchmark remains a static MalImg image setting. Further work can therefore examine whether a different structural representation improves class-wise robustness and whether explanations/error patterns remain reliable across models. | A very recent high-performing MalImg benchmark. It is useful for establishing how close a proposed method is to current CNN/ensemble performance rather than comparing only with older baselines. |

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| 5 | Alotaibi, B. (2025). Multimodal Deep Learning Fusion for Accurate and Explainable Malware Family Classification. Applied Sciences, 15(21), 11635. | Applied Sciences (MDPI) 2025 SJR 2025: Q2; JCR 2025: Q2 | MalImg + Fusion Malware; ConvNeXt-Tiny visual features + static features; cross-attention, snapshot ensembling, TTA, per-class calibration, and XAI. | This recent work already addresses explainability, class imbalance, calibration, and multimodal fusion, showing that aggregate accuracy can hide difficult families. However, its MalImg component still relies on visual/static feature fusion rather than explicitly representing spatial relationships as a graph. The remaining challenge is to determine whether a graph-structured representation adds useful information beyond strong CNN and multimodal baselines, especially for difficult or underrepresented families. | Provides a strong 2025 benchmark with explainability and macro-F1 considerations. It supports a research direction focused on structural representations and deeper error analysis rather than simply seeking a higher accuracy number. |

**References**

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